

Impact of Repeated Impulse Voltage on the Breakdown and Space Charge Characteristics of Polypropylene for High-Voltage Cable Insulation

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Abstract—Polypropylene (PP) exhibits excellent electrical insulation properties, recyclability, and high-temperature stability, making it a eco-friendly material for high-voltage cable insulation. Repeated impulse voltages (such as those generated by lightning strikes and switching operations) may cause degradation of polypropylene insulation and compromise its reliability. This study investigates the breakdown characteristics and space charge accumulation patterns of polypropylene insulation materials under repeated impulse voltages. The 50% impulse breakdown voltage of the material was measured using the step-up method, and an exponential relationship model was established between the amplitude of the impulse voltage and the average number of pulses required for breakdown to occur. Pulse electroacoustic (PEA) testing and analysis were conducted to examine the space charge distribution characteristics of the material under a direct current high-field environment after exposure to different numbers of impulse voltages (0, 50, 100, 200 times). The results indicate that when the number of impulse voltages reaches 100 or more, significant space charge accumulation occurs near the cathode inside the material, and this phenomenon further intensifies when the number of impulse voltages increases to 200. The study reveals a significant correlation between the number of impulse voltages and space charge accumulation, providing experimental evidence and references for optimizing the impact resistance of polypropylene insulating materials.

Keywords—High-Voltage Cable, Polypropylene, Impulse Voltage, Space Charge, Insulation Degradation

I. INTRODUCTION

High-Voltage Cables play an indispensable role in modern power systems. During actual operation, cable insulation is not only subjected to power frequency electrical stress but also experiences short-term impulse electrical stress due to factors such as lightning overvoltage, operational overvoltage, and resonant overvoltage. For example, between 1999 and 2008, the annual maximum number of lightning strikes in Guangdong Province reached 112,106, with transmission

lines suffering nearly 400 lightning strikes [1]. These pulse voltages may affect the long-term reliability of cable insulation materials, which is an area of concern.

W. Sima et al. found that the cumulative effect of multiple impulse voltages can increase key dielectric factors such as volume conductivity, dielectric constant, and dielectric loss of oil-paper insulation [2]. N. PAMUK et al. conducted insulation degradation analysis on XLPE cable insulation materials under switching pulse voltages and found that partial discharge signals increased significantly after the application of switching pulse voltages [3].

Polypropylene (PP) is a thermoplastic material that is easy to recycle [4] and does not require cross-linking. It also has a high melting point and a high rated operating temperature. These advantages make it a promising candidate to replace cross-linked polyethylene (XLPE) as a new generation of environmentally friendly electrical cable insulation material. However, repeated lightning strikes can cause degradation and deterioration of polypropylene, and there is currently little research on this topic.

Based on this, this paper uses the step-up method to test the impulse breakdown voltage of polypropylene, establishing a statistical relationship between the impulse voltage amplitude and the average number of pulses required for breakdown to occur. It also tests the space charge of polypropylene under 50, 100, and 200 impulse cycles, studying the charge accumulation characteristics. The research findings provide experimental evidence for the insulation status of polypropylene insulated cables under repeated impulse voltages.

II. EXPERIMENTAL METHODS

A. Materials and Sample Preparation

The polypropylene matrix material studied in this paper is Yanshan Petrochemical 8003, and the elastomer is based on POE (polyolefin elastomer) with a mix ratio of 60:40. Sheet

samples were prepared from polypropylene granules. The polypropylene granules were placed in a high-temperature mixer and mixed for 5 minutes, then hot-pressed on a flat vulcanizing machine. The hot-pressing temperature was 190°C, and the samples were hot-pressed for 3 minutes at 5 MPa, 10 MPa, and 20 MPa, respectively. The molded samples were annealed in a vacuum oven at 100°C, with a cooling rate of 0.1 K/min and a vacuum oven vacuum pressure of 0.1 Pa. The final samples had dimensions of 10 cm × 10 cm and a thickness of 200 μm. The process and sample parameters are shown in Fig. 1.

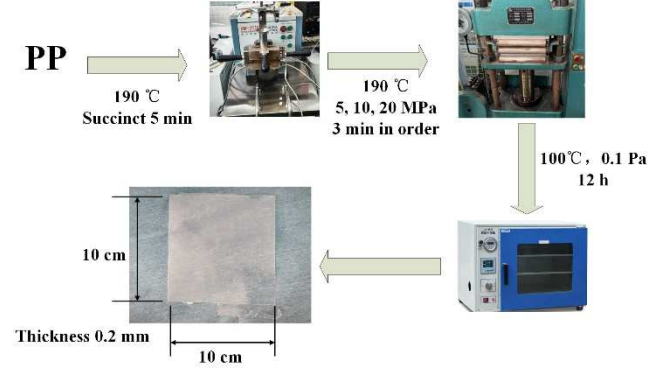


Fig. 1. Polypropylene sample preparation procedure

B. Impulse Voltage Breakdown System

The impulse voltage generator used in this test is shown in Fig. 2. It can generate an impulse voltage of up to 400 kV through a four-stage configuration. The multi-stage impulse voltage generator utilizes the principles of parallel charging and series discharging. The capacitors in each stage are instantly connected in series to release voltage, forming an impulse voltage with a high amplitude.

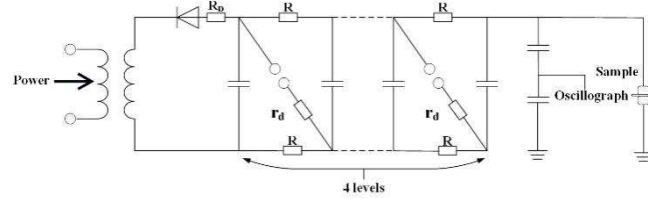


Fig. 2. Multi-stage impulse voltage generator

By adjusting the front and rear resistors to modify the discharge time constant, a waveform that meets the system test requirements is obtained, as shown in Fig. 3. The rise time is approximately 300 ns, and the half-peak time is approximately 50 μs.

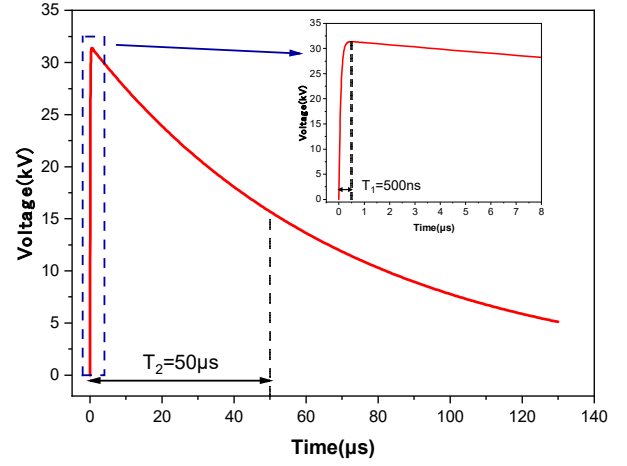


Fig. 3. Impulse voltage waveform

C. Space Charge Testing System

The basic principle of measuring the spatial charge distribution in this paper is the pulsed electro-acoustic method (PEA). The spatial charge measurement system is shown in Fig. 4, where the pulse power supply outputs pulses with a width of 5 ns, a repetition frequency of 50 Hz, and an output pulse amplitude of 0–2 kV. The piezoelectric sensor consists of a 9 μm polyvinylidene fluoride (PVDF) film.

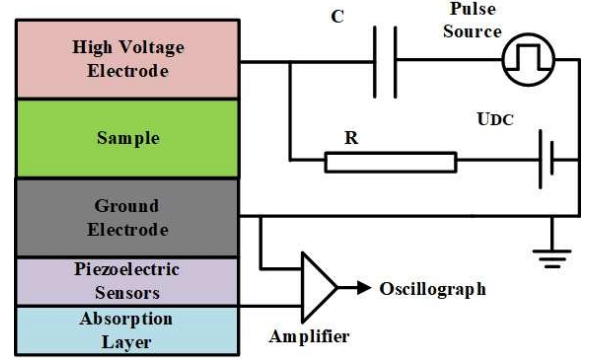


Fig. 4. Space charge measurement system

III. RESULTS AND DISCUSSION

A. Impulse Voltage Breakdown Characteristics

According to IEC 62271-200:2011[5], the step-up method was used to study the impulse breakdown voltage of polypropylene insulating materials. Based on the previous AC breakdown test conducted on a 200 μm polypropylene sample, the impulse breakdown field strength of polypropylene insulating materials was preliminarily estimated to be between 150 kV/mm and 250 kV/mm. Therefore, in this step-up test, the sample temperature when the repeated pulse voltage is applied is 20°C, the initial voltage was set to 34 kV, with adjustments of ±2 kV made based on whether the sample underwent breakdown. The test data obtained are shown in Table I:

TABLE I. RESULTS OF POLYPROPYLENE BREAKDOWN TEST

Voltage/kV	n	$\sum V \cdot n$
34.4	1	34.4
36.4	2	72.8
38.4	4	153.6
40.4	5	202
42.4	3	127.2

$$U_{50\%} = \frac{\sum \sum V_{in}}{N} \quad (1)$$

Calculated 50% impulse breakdown voltage of polypropylene insulating material is $U_{50\%} = 40.29$ kV. In the engineering operation, the actual situation is usually faced with a number of lower amplitude impulse voltage. Therefore, this paper to 50% impulse breakdown voltage as a benchmark, to 2 kV as a gradient to continuously reduce the voltage amplitude, the time interval between each impulse for 30 s, measurement of the impulse voltage amplitude U and make the specimen breakdown of the average number of impulse voltage required N . The average number of times required to break down the average impulse voltage amplitude of different impulse voltages to do an exponential fit. An exponential function with e as the base is used for the fitting:

$$y = y_0 + A_1 e^{-\frac{x}{t_1}} \quad (2)$$

Fig. 5 shows the average number of impulse voltages as well as the results of exponential fitting, the confidence coefficient R^2 of the fit is 0.9995, which indicates that the model fits this set of data extremely well. Therefore, it is concluded that the exponential model can explain the breakdown characteristics of polypropylene insulation under multiple shock voltages. The equation obtained from the fitting is shown in the following:

$$y = 1.3 \times 10^{11} \times e^{-\frac{x}{1.7}} + 0.2 \quad (3)$$

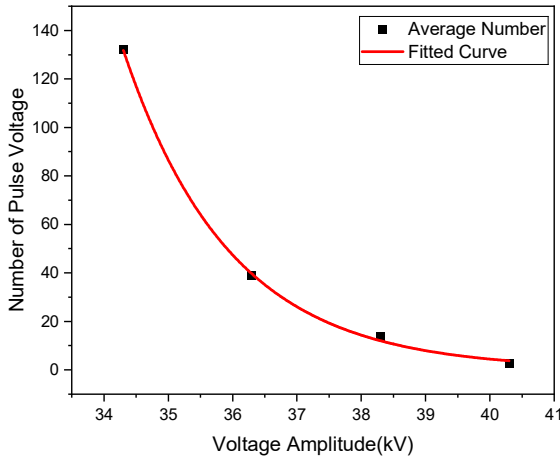


Fig. 5. U-N characteristics of polypropylene at impulse voltage

B. Space Charge Characteristics

To further study the effect of impulse voltage on the dielectric properties of polypropylene and to evaluate the space charge accumulation characteristics of insulating materials subjected to an applied DC high field after multiple impulse voltages, this paper takes the impulse voltage with an amplitude of 32 kV as a reference, applies 0, 50, 100 and 200 times of the impulse voltage to polypropylene specimens and observes the internal space charge distribution of polypropylene under the action of DC voltage. The temperature of the specimen was maintained at 30°C during the test.

In order to exclude the effect of surface induced charges after the impulse voltage, all specimens were tested after 8 h of static short-circuiting after the impulse voltage. The space charge distributions of the intact specimens that have not been

subjected to multiple impulse voltages and the specimens that have been subjected to 50 impulse voltages are shown in Fig. 6 and Fig. 7, respectively. It can be visualized that in both types of specimens, there is almost no significant space charge distribution inside the polypropylene specimens after 30 min of DC high voltage (20 kV/mm) pressurization. This indicates that the 50-shock voltage has no significant effect on space charge buildup in the insulating material, and that the space charge characteristics are almost identical to those of the intact specimens. From the point of view of space charge accumulation, this result shows that the electrical properties of the insulating material have not yet been significantly degraded by a smaller number of impulse voltages.

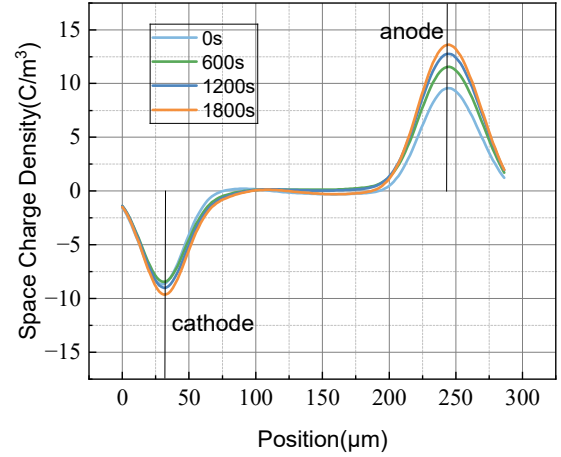


Fig. 6. Space charge distribution before applied impulse voltage

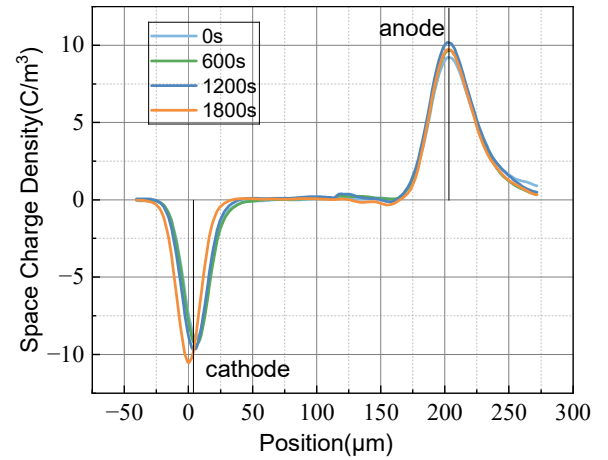


Fig. 7. Space charge distribution after 50 times impulse voltages

The space charge distribution inside the polypropylene specimen is further investigated for 100 and 200 times of impulse voltage, and the results are shown in Figs. 8 and 9, respectively. A large amount of space charge exists near the cathode with the increase of the pressurization time, and there are more and more negative charges of the same polarity near the cathode with the increase of the pressurization time. When the pressurization time reaches 1800 s, about 5 C/m³ of space charge accumulates.

As the number of externally applied impulse voltages increases, the negative charge buildup near the cathode inside the specimen becomes more and more serious. After the polypropylene specimens were subjected to 50 impulse voltages, the space charge suppression ability of the specimens could still be maintained at a high level at this time.

There was almost no negative charge accumulation inside the specimen after 1800 s of voltage application. When the number of impulse reaches 100 times, after 1800 s of DC high voltage, there is a significant space charge distribution inside the specimen, and the maximum value of space charge density is about 5 C/m^3 . When the number of shocks reaches 200, the absolute value of space charge density at each position increases by about 2 C/m^3 .

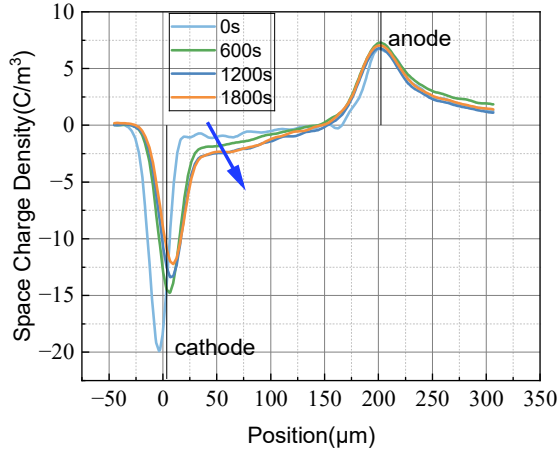


Fig. 8. Space charge distribution after 50 times impulse voltages

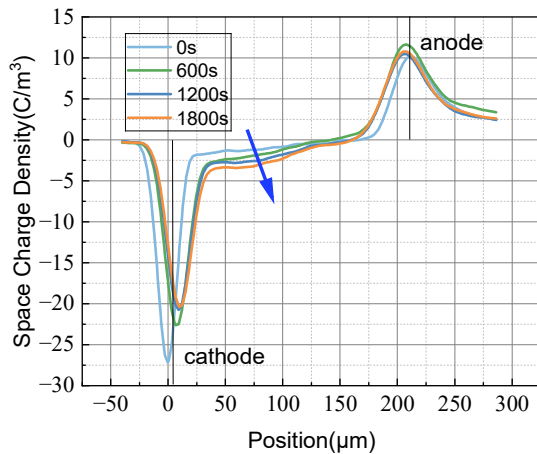


Fig. 9. Space charge distribution after 50 times impulse voltages

Under the continuous action of long-term shock voltage, a large number of carriers are first injected into the polypropylene insulator through the Schottky effect [6]; the electrical stress prompts the polymer chain cleavage, and the generated small-molecule by-products are both a new source of space charge, and may also become a trap of varying depths, altering the charge capture-release kinetics. At the initial stage of impulse, the impurity ions formed by the dissociation of by-products are mainly enriched, and the space charge is mostly confined to the region adjacent to the electrodes; with the increase of the number of impulse, the interior of the specimen gradually exhibits the carrier injection and migration behaviors similar to those under the DC electric field, and the charge cloud expands to the depth of the volume, which highlights the far-reaching influence of the long-term impulse voltage on the internal charge transport channel. At the same time, the energetic charged particle bombardment and local heat accumulation generated during the discharge process

continuously destroy the polypropylene chain segments and induce thermal cracking, and the structural damage and space charge effect are superimposed on each other, resulting in a nonlinear decay of dielectric pressure resistance and thermal stability with service time. From a comprehensive point of view, the impulse voltage changes the insulation performance by both direct introduction of space charge and progressive structural degradation, and its coupling mechanism and evolution law still need to be further studied systematically to support the design of high-reliability insulation.

IV. CONCLUSION

This study examines the breakdown behaviour and space-charge accumulation of elastomer-modified PP insulation under repeated impulse voltages. The conclusions are as follows:

(1) For a 200 μm -thick specimen, the 50 % impulse breakdown voltage is 40.29 kV , and the impulse amplitude shows a pronounced exponential correlation with the mean breakdown count, yielding an excellent fit ($R^2 = 0.9995$).

(2) After different numbers of impulses, the space-charge accumulation under a subsequent DC high field varies markedly. With fewer than 50 impulses, the charge profile remains essentially stable; at 100 impulses, pronounced accumulation appears near the cathode, where the charge density exceeds 5 C/m^3 after 1800 s; after 200 impulses, the charge density in all regions rises by about 2 C/m^3 .

(3) Impurity ions generated in the initial stage aggregate near the electrodes; as the impulse count rises, carrier injection becomes evident and bulk space-charge build-up intensifies.

These results reveal a clear association between impulse repetitions and space-charge accumulation, deepen the understanding of the underlying mechanisms, and provide valuable guidance for optimising PP insulation structures and enhancing their resistance to impulse stress.

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